

B. LIPIKA

AI/ML Engineer | Generative AI | Industrial Analytics
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Professional Summary

AI/ML engineer with hands-on experience building industrial time-series analytics, forecasting, anomaly-detection, and LLM-powered data workflows. Skilled in Python, SQL, PyTorch, LangGraph, FastAPI, and local LLMs; designs evidence-driven systems that separate deterministic computation from LLM reasoning.

Technical Skills

Programming: Python, SQL, Java
Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, LSTM, SHAP
Time-Series: Forecasting, trend/seasonality analysis, anomaly and changepoint detection, rolling statistics
Generative AI: LangGraph, LangChain, Ollama, local LLMs, NLP, AI agents, MCP, tool calling
Data & Backend: Pandas, NumPy, PyArrow, DuckDB, Parquet, FastAPI, Flask, REST APIs, SQL
Tools: Git, GitHub, Docker, Jupyter, Postman, VS Code

Experience

AI/ML Trainee Jan 2026 – Present
Tiny Prism Labs, Manipal

- Building an industrial analytics assistant that translates natural-language questions into sensor-data retrieval, analysis, and visualization workflows.
- Developed Python query-understanding and Flask API components, plus schema-aware tools for dataset inspection, SQL retrieval, and time-series analysis.
- Integrated local LLMs into evidence-driven workflows for trend, seasonality, anomaly, changepoint, and correlation analysis, keeping calculations deterministic.

Projects

Industrial Time-Series Forecasting GitHub
Python, PyTorch, IBM Granite TTM-R3, FastAPI
Fine-tuned/evaluated a time-series model for 15-minute industrial inflow forecasts using chronological splits, error metrics, and spike-focused error analysis; deployed a rolling-forecast API.

Industrial AI Agent Platform GitHub
Python, LangGraph, local LLMs, SQL, FastAPI
Built a modular Router–Analytics–Reasoning workflow with schema-aware tools for dataset inspection, SQL retrieval, time-range discovery, and deterministic sensor analytics.

AI Data Scientist Agent GitHub
Python, Pandas, Scikit-learn, local LLMs
Created an AI-assisted data-science workflow for dataset profiling, schema discovery, analytical tool selection, and ML experimentation, grounding LLM interpretations in computed results.

Industrial Vibration Anomaly Detection GitHub
Python, Pandas, Parquet, Isolation Forest
Analyzed 2.3M+ industrial vibration records with data-quality checks, statistical exploration, and Isolation Forest to identify unusual sensor behavior and visualize anomalies.

Industrial Sensor Data Lake Pipeline GitHub
Python, DuckDB, Parquet, PyArrow, SQL
Built an incremental sensor-data pipeline with validation, transformation, partitioned Parquet storage, hourly/daily rollups, and a natural-language interface for validated SQL analytics.

Technical Exposure

NXP Semiconductors: Time-series/edge AI and eIQ Time Series Studio
Electronica India 2026, Bengaluru: Industrial AI, edge computing, and embedded technologies.

Innovation & Achievements

AgriFlow – Smart Irrigation: Indian utility patent holder (No. 202541061616); developed sensor-based soil monitoring and automated irrigation control.

Retina AI: Led development of a computer-vision assistive system using YOLOv8/OpenCV with tactile vibration and voice feedback.

Recognition: 2nd Runner-Up, VTU Mini Project Competition | Google Data Analytics Professional Certificate (Coursera, 2025) | Java Programming (NPTEL, 2024)

Education

B.E., Information Science and Engineering 2022 – 2026
Canara Engineering College, Mangalore (VTU) CGPA: 8.25/10